EXPERIMENT-2

Aim:

Convert the ER diagram into a relational scheme. Write complete MY SQL CREATE TABLE statements with Primary Keys, Foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

```
CREATE TABLE Customer (
        CustomerID INT AUTO_INCREMENT PRIMARY KEY,
        Name VARCHAR (100) NOT NULL,
        Email VARCHAR (100) NOT NULL UNIQUE,
        Phone VARCHAR (15)
        UNIQUE (Phone)
) ENGINE = InnoDB ;

CREATE TABLE Address (
        AddressID INT AUTO_INCREMENT PRIMARY KEY,
        CustomerID INT NOT NULL,
        TYPE ENUM ('Home','Work') NOT NULL,
        Street VARCHAR (200) NOT NULL.
        City VARCHAR(100) NOT NULL,
        State VARCHAR (100) NOT NULL.
        Pincode VARCHAR (10) NOT NULL.
        FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
                ON DELETE CASCADE ON UPDATE CASCADE
) ENGINE = InnoDB;
```

```
CREATE TABLE Fashion (
        ProductID INT PRIMARY KEY.
        Size VARCHAR(20).
        Color VARCHAR(20),
        FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
            ON DELETE CASCADE
) ENGINE = InnoDB;

CREATE TABLE Books (
        ProductID INT PRIMARY KEY.
        Author VARCHAR(100),
        ISBN VARCHAR(20) UNIQUE,
        FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
            ON DELETE CASCADE
) ENGINE = InnoDB;

CREATE TABLE Order (
        OrderID INT AUTO_INCREMENT PRIMARY KEY,
        CustomerID INT NOT NULL,
        OrderDate DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
        TotalAmount DECIMAL(12,2) NOT NULL.
        FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
            ON DELETE CASCADE ON UPDATE CASCADE
) ENGINE = InnoDB;

CREATE TABLE OrderItem (
        OrderID INT NOT NULL,
        ProductID INT NOT NULL.
        Quantity INT NOT NULL CHECK(Quantity >0),
        UnitPrice DECIMAL(10,2) NOT NULL,
        Discount DECIMAL(5,2) DEFAULT0,
        PRIMARY KEY (ProductID, OrderID).
```

```
        FOREIGN KEY (OrderID) REFERENCES 'Order' (OrderID)
            ON DELETE CASCADE ON UPDATE CASCADE,
        FOREIGN KEY (ProductID) REFERENCES Product (ProductID)
            ON DELETE RESTRICT ON UPDATE CASCADE
) ENGINE = InnoDB;


CREATE TABLE Payment (
        PaymentID INT AUTO-INCREMENT PRIMARY KEY,
        OrderID INT NOT NULL UNIQUE,
        Amount DECIMAL (12,2) NOT NULL,
        Method ENUM ('UPI','Card','COD','NETBANKING') NOT NULL,
        PaymentDate DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
        TransactionID VARCHAR (100) UNIQUE,
        FOREIGN KEY (OrderID) REFERENCES 'Order' (OrderID)
            ON DELETE CASCADE ON UPDATE CASCADE
) ENGINE= InnoDB;
```

```
CREATE TABLE Delivery (
        DeliveryID INT AUTO-INCREMENT PRIMARY KEY,
        OrderID INT NOT NULL UNIQUE,
        DeliveryDate DATETIME,
        TrackingNo VARCHAR(100) UNIQUE,
        Status ENUM('Pending', 'Shipped', 'Out for Delivery', 'Delivered')
            DEFAULT 'Pending',
    FOREIGN KEY (OrderID) REFERENCES 'Order' (OrderID)
        ON DELETE CASCADE ON UPDATE CASCADE
) ENGINE=InnoDB;
```

Insert Sample Data

```
INSERT INTO Customer (Name, Email, Phone)
VALUES ('Aarav Sharma', 'aarav@gmail.com', '9876543120');

INSERT INTO Address (CustomerID, Type, Street, City, State, Pincode)
VALUES (1, 'Home', '12 MG Road', 'Mumbai', 'Maharashtra', '400001');

INSERT INTO Seller (SellerName, Email, Phone)
VALUES ('TechBazaar', 'info@techbazaar.in', '9000000001');

INSERT INTO Category (CategoryName, Descriptions)
VALUES ('Electronics', 'Mobiles, Laptops, Accessories');

INSERT INTO Product (ProductName, Price, Stock, CategoryID, SellerID)
VALUES ('Smartphone', 15999.00, 1, 1);

INSERT INTO Electronics (ProductID, WarrantyPeriod)
VALUES (1, 12);

INSERT INTO 'Order' (CustomerID, TotalAmount)
VALUES (1, 15999.00);

INSERT INTO OrderItem (OrderID, ProductID, Quantity, UnitPrice)
VALUES (1, 1, 1, 16599.00);

INSERT INTO Payment (OrderID, Amount, Method, TransactionID)
VALUES (1, 15999.00, 'UPI', 'TXN123456');

INSERT INTO Delivery (OrderID, DeliveryDate, TrackingNo, Status)
VALUES (1, NOW(), 'TRX123456', 'Shipped');
```

REFERENTIAL INTEGRITY VIOLATIONS

```
--1) Inserting OrderItem with invalid ProductID
INSERT INTO OrderItem (OrderID, ProductID, Quantity, UnitPrice)
VALUES (1, 999, 1, 1000.00);
```

ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraints fails (ecommerce, OrderItem, CONSTRAINT 'orderitem_ibfk_2', FOREIGN KEY ('ProductID') REFERENCES 'Product' ('ProductID'))

-- Deleting Customer referenced by Order (should fail due to RESTRICT)

```
DELETE FROM Customer WHERE CustomerID=1;
```

~~Error~~

ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`ecommerce`.`Order`, CONSTRAINT `Order_ibfk_1` FOREIGN KEY (`CustomerID`) REFERENCES `Customer` (`CustomerID`))

Output

Customers

CustomerID	Name	Email
1	Aarav Sharma	aarav@gmail.com

OrderItem

OrderID	ProductID	Quantity	UnitPrice
1	1	1	1

Delivery

DeliveryID	OrderID	Status	TrackingNo
1	1	Shipped	TRK123456